

Particle swarm optimization

Particle swarm optimization is a method which simulates the movement of many particles, which tend to the minimum of the goal function.

At the beginning we initialize each particle with a random position and velocity. We store the best position each particle has had since the beginning and the known best global position, of course everything should be updated every step.

At the end of each step we add particle's velocity to its position and update its velocity according to the formula:

$$v_i(t+1) = w * v_i + r_g * c_g * (x_i - g) + r_p * c_p * (x_i - p_i)$$

where

- w, c_g, c_p – simulation parameters
- g – known best global position
- p_i – particle's known best position
- v_i – particle's velocity
- x_i – particle's current position
- r_g, r_p – random numbers $<0;1>$

1. Use particle swarm optimization to find the global minimum of the function:

$$f(x,y) = \sin^2(x) + \sin^2(y) + \sin(x) \sin(y) \quad x,y \in [-3;3]$$

2. Try to implement the discrete version of this approach to solve the Partition problem.